Curiosity-Enhanced Cooperative Trajectory Planning Algorithm for Multi-UAV Data Collection

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***Abstract*—Unmanned aerial vehicles (UAVs) have been utilized in data collection to improve the efficiency of wireless sensor networks (WSNs) due to their flexible mobilities. The trajectories of UAVs affect the efficiency of WSN and energy consumption. However, UAV-enabled data collection in complex and dynamic environments presents significant challenges. This paper addresses the trajectory planning for multi-UAV-enabled data collection in complex and dynamic environments. In this work, a curiosity-enhanced cooperative trajectory planning (CCTP) algorithm based on deep reinforcement learning (DRL) is developed. A deep recurrent Q-network (DRQN) that aggregates historical input information is deployed for trajectory planning in partially observable and dynamic environment. Next, a dual-incentive reward mechanism that integrates extrinsic environmental reward and intrinsic curiosity reward is proposed for efficient policy training. An intrinsic curiosity module (ICM) is designed to endogenously generate the curiosity reward. Numerous simulation results are presented to show that UAVs exhibit shorter completion time and higher data collection rate using proposed algorithm than other comparative benchmarks.**

***Index Terms*—trajectory planning, unmanned aerial vehicle (UAV), data collection, deep recurrent Q-network (DRQN), curiosity reward.**

I. Introduction

Unmanned aerial vehicles (UAVs) effectively facilitate data collection for Internet of Things (IoT) devices in inaccessible areas where the qualities of ground-based channels are poor, such as forests, oceans and disaster areas. The timeliness of data collection and the energy consumption of UAV are the major issues of UAV-enabled data collection. By optimizing the collection node visitation sequence and flight route, the path planning of UAV has effects on the collection time and energy consumption.

In recent years, the UAV path planning for data collection was studied by developing traditional optimization algorithms, nature-inspired algorithms and deep reinforcement learning (DRL). Traditional optimization algorithms solve waypoints for data collecting by optimization functions, such as convex optimization. This kind of methods pre-plans trajectories based on deterministic and complete environmental information before collection task begins [2], [3], [15]. Nature-inspired algorithms search the optimal trajectory from solution space by mechanisms that emulate the principles of nature, which effectively solve large-scale data collection and multi-UAV collection scenarios [4]. DRL-based algorithms train the neural network models for trajectory planning by the experience from trail and error experiments. These DRL models make trajectory planning actions based on real-time environment states, thereby adapting to dynamically changing environments. However, the existing works have the following limitations:

Firstly, the reliance on the prior and global knowledge of the environment. Some works pre-plan trajectories based on the prior environmental information [2]–[4], [15]. Li et al. proposed an algorithm that optimizes hovering points for data collection and connects them to form a flight route [15]. The work [4] proposed a trajectory planning approach that searches the optimal collection sequence using nature-inspired algorithm. These methods are available in environments that are fixed, and their information is known in advance. However, in post-disaster communication scenario, the node information is unknown in prior. Besides, in some scenarios such as geological and ocean monitoring, the locations of sensor nodes will unpredictably change. Some works proposed path planning methods for dynamically changing environments. Wang et al. [5] proposed a heuristic path planning algorithm based on dynamically adjusting regional coordination for collection nodes. When the environment is dynamically changing, real-time planning methods such as DRL are more suitable [6], [7]. For example, Dai et al. [6] proposed a DRL-based algorithm for CCTV data collection, where collection nodes dynamically generate data which need to be collected as soon as possible. Despite the advantage in dynamic environment, these algorithms still necessitate the complete information of the environmental state to generate a viable trajectory. The real-time information of the environmental state is usually obtained by airborne sensors, but sensors have limited sensing range in vast environment. Therefore, overcoming this dependency on global information remains a critical challenge for trajectory planning in dynamic and partially observable environments.

Secondly, the obstacle constraints in trajectory planning, especially the obstacles with mobility. The majority of trajectory planning studies for data collection focus on optimizing wireless communication channels, neglecting the obstacle constraint on UAV flight [2]–[4], [7], [8]. In most scenarios, the UAV flights are impacted by static obstacles (e.g., terrain, buildings) as well as dynamic obstacles (e.g., birds, irrelevant air vehicles). Xie et al. [7] studied the real-time

traffic based vehicular data collection, but failed to consider the impact of obstacles in urban area, such as tall buildings. Thus, it is essential to take obstacle constraints into account. There are two kinds of methods through which UAV agent recognize the constraint of obstacles: one is the fixed learning of obstacle locations [9], and the other is encoding the real-time obstacle sensory data [6], [10]. The algorithm proposed in [9] is designed to learn the locations where collisions with obstacles would occur. The collision location memory in this algorithm is fixed. When the UAV is moved to another environment, the trained algorithm is unavailable. On the contrary, the algorithms that take real-time obstacle sensory data as input demonstrate strong generalization across diverse environments. In real-time obstacle avoiding, the algorithms [6], [11] describe the environment information by grid maps. A limitation of these algorithms is that the generation of state map rely on global sensory data which cannot be directly measured by airborne sensors (e.g., LiDARs).

To overcome these limitations of prior work, in this paper, a curiosity-enhanced cooperative trajectory planning (CCTP) algorithm for multiple UAV data collection in complex and dynamic environment is proposed. The main contributions are as follows:

- A DRL model with recurrent mechanism is developed for trajectory planning in partially observable environment. The policy makes a precise control action based on aggregating real-time and historical environmental state information.
- A dual-incentive reward mechanism that integrates extrinsic environmental reward and intrinsic curiosity reward is proposed for efficient policy training. The curiosity reward promotes comprehensive exploration of environment during the training phase, thereby facilitating the learning of the optimal policy in complex environment.
- Numerous simulation experiments are conducted to evaluate the performances of proposed algorithm and benchmark algorithms. The results show that the proposed CCTP algorithm achieves a significant reduction in completion time while ensuring the collection of the most of data.

II. Preliminaries

A. System model

In this work, a multi-UAV data collection for forest monitoring sensors scenario is considered. The dispatched UAVs plan their optimal flying routes to collect the data from all the sensor nodes and avoid any conflict between UAVs. Besides, UAVs have to navigate through obstacles like tall trees that are randomly distributed, especially some of these obstacles exhibit mobility, such as bird flocks. The system contains N UAVs which are denoted as $\mathcal{U} = \{U_1, U_2, \cdots, U_N\}$. Each UAV flies at a fixed height H with constant velocity V. Hence, the position of UAV U_i is denoted as $\mathbf{p}_{U_i} = (x_i, y_i, H)$. The controlling variable of UAV is its steering angle. The IoT sensor nodes are randomly distributed on the ground. This system contains M IoT sensor nodes that are denoted as $\mathcal{I} = \{I_1, I_2, \cdots, I_M\}$. The position of node I_j is denoted as $\mathbf{p}_{I_j} = (x_j, y_j, 0)$. The IoT sensor node I_j stores random amount of data that is denoted as D_{I_j}. In this system model, the set of obstacles is denoted as $\mathcal{B}$.

When the UAV is in the communication range of IoT sensor node, the air-to-ground communication link will be established to transmit the stored data to the UAV. As affected by the barriers like trees and buildings, the communication channel is modelled as statistical mixed channel of both line-of-sight (LoS) channel and non-line-of-sight (NLoS) channel in urban area [22]. The probability of existence of LoS channel relevant to the elevation angle of UAV and IoT node, i.e.

$$P_{LoS}^{i,j} = 1/\left(1 + \alpha e^{-\beta(\theta_{i,j} - \alpha)}\right), \tag{1}$$

where $\theta_{i,j}$ is the angle of elevation between the UAV U_i and the IoT sensor node I_j and α, β are the environmental parameters. Hence, the probability of NLoS link is $P_{NLoS}^{i,j} = 1 - P_{LoS}^{i,j}$. The path loss of the communication channel is regarded as the expectation of the occuring of LoS and NLoS, i.e.

$$\bar{\Lambda}_{i,j} = K_0 d_{i,j}^2 \left(P_{LoS}^{i,j} \cdot \mu_{LoS} + P_{NLoS}^{i,j} \cdot \mu_{NLoS}\right), \tag{2}$$

where $K_0 = (4\pi f_c d_0/c)^2$, f_c is the frequency of carrier, c is the propagation velocity of light, d_0 is the reference distance, $d_{i,j}$ is the distance between the UAV U_i and the IoT sensor node I_j, μ_{LoS} and μ_{NLoS} are the decaying factors corresponding to the LoS link and NLoS link. The reference distance d_0 is considered as 1m. In this signal transmission model, the data rate for the transmission between the UAV U_i and the IoT sensor node I_j is denoted as:

$$R_{i,j} = B \log_2 \left(1 + \frac{P_{I_j}}{\sigma^2 \bar{\Lambda}_{i,j}}\right), \tag{3}$$

where B is the bandwidth, P_{I_j} is the signal transmission power of I_j. The UAV hovers above the IoT node and receives the data with rate $R_{i,j}$.

B. Problem formulation

To improve the efficiency of data collection, the proposed algorithm is expected to help UAVs finish the data collection task with the shortest time and maximum collected data. The task completion time T is defined as starting from the UAVs leave the starting point and ending when the latest UAV arrives at the destination. In data collection process, the data transmission starts when the UAV flies into the communication range $d_{\max}^I$. The communication state of IoT sensor node at time slot t is denoted as:

$$\varphi_t^{i,j} = \begin{cases} 1, & \|\mathbf{p}_t^{U_i} - \mathbf{p}_{I_j}\| < d_{\max}^I \\ 0, & \|\mathbf{p}_t^{U_i} - \mathbf{p}_{I_j}\| > d_{\max}^I \end{cases}, \tag{4}$$

where $\varphi_t^{i,j} = 1$ indicates that the IoT sensor node I_j is transmitting data to the UAV. The amount of collected data from all the IoT sensor nodes is denoted as:

$$D = \sum_{I_j \in \mathcal{I}} \sum_{U_i \in \mathcal{U}} \sum_{t \in T} R_t^{i,j} \varphi_t^{i,j} \Delta t, \tag{5}$$

where Δt is the length of each time slot. Consequently, as $\mathbf{P}_U$ is the set of trajectories of all the UAVs, the trajectory optimization on the maximum collected data and minimum task completion time is denoted as:

$$\begin{aligned} P1: &\max_{\mathbf{P}_U} (\omega_1 D - \omega_2 T) \\ s.t.\ C1: &D_c^{I_j} \leq D_{I_j},\ \forall I_j \in \mathcal{I}, \\ C2: &d_{i,m} > d_{\min}^B, \forall U_i \in \mathcal{U}, \forall B_m \in \mathcal{B}, \\ C3: &d_{\min}^U < \min\left(d_{i,k}^U\right) < d_{\max}^U, \forall U_i, U_k \in \mathcal{U}, U_i \neq U_k, \end{aligned} \tag{6}$$

where ω_1, ω_2 are weighting factors, $D_c^{I_j}$ is the amount of collected data at IoT sensor node I_j, $d_{i,m}$ is the distance between the i-th UAV and the surface of obstacle B_m, $d_{\min}^B$ is the minimum distance to the obstacle surface for safety, $d_{i,k}^U$ is the distance between the UAV U_i and the UAV U_k. $d_{\min}^U$ is the minimum distance for safety and $d_{\max}^U$ is the maximum distance between UAVs. Constraint $C1$ limits the amount of collected data will not exceed the amount of stored data, constraint $C2$ requires UAVs to fly around obstacles and $C3$ requires UAVs to avoid collision with the nearest adjacent UAV while keeping a certain distance to each other to maintain the air-to-air communication.

III. Proposed Algorithm

A. Modelling data collection trajectory planning as POMDP

The trajectory planning for data collection in this forest environment faces these two challenges: (1) The distributions of IoT sensor nodes and obstacles are unknown for UAVs before the data collection episode starts, and UAVs rely solely on localized sensory data due to limitations in onboard perception capabilities. (2) The environment undergoes dynamic variations due to the presence of moving obstacles and concurrent data collection operations by multiple UAVs. Therefore, it is more appropriate to model the trajectory planning problem as partially observable Markov decision process (POMDP), where UAVs are modelled as an agent that execute trajectory planning through a sequential decision-making mechanism. Each agent acquires the observation of environment state through localized sensing and selects the optimal trajectory controlling action based on the observation.

(a) Observations: The set of observations from all the agents is denoted as $\mathcal{O} = \{o_1, o_2, \cdots, o_N\}$. For finishing data collection task with other UAVs, each agent observes the state of destination, IoT sensor nodes, obstacles and adjacent UAVs.

- *Destination-related information:* To find a path to the destination, the agent observes the state of destination, i.e. $o_{tar} = (d_{tar}, \theta_{tar})$, where d_{tar} is the distance to the destination, θ_{tar} is the direction angle to the destination.
- *Observed IoT sensor nodes:* To find the optimal path to IoT nodes, the agent observes the states of 5 nearest IoT node, i.e. $o_I = \left\{\left(id_1^I, d_1^I, \theta_1^I, D_1^I\right), \cdots, \left(id_5^I, d_5^I, \theta_5^I, D_5^I\right)\right\}$, where id_j^I is the identical number of the j-th nearest node, d_j^I is the distance to the node, θ_j^I is the direction angle of node based on the UAV's heading direction, and D_j^I is the data amount of the node in bytes.
- *Detected obstacles:* To obtain the information of obstacles, the agent observes the obstacles by range finder that has 16 detection directions, i.e. $o_B = (d_1, d_2, ..., d_{16})$, where d_i is the distance to the surface of obstacle in the i-th direction.
- *Adjacent UAVs:* To keep the distance with each adjacent UAV, the UAV observes 2 nearest adjacent UAVs, i.e. $o_U = \left(d_1^U, \theta_1^U, d_2^U, \theta_2^U\right)$, where d_k^U is the distance to the k-th nearest UAV, θ_k^U is the direction angle of the adjacent UAV based on the local UAV's heading direction.

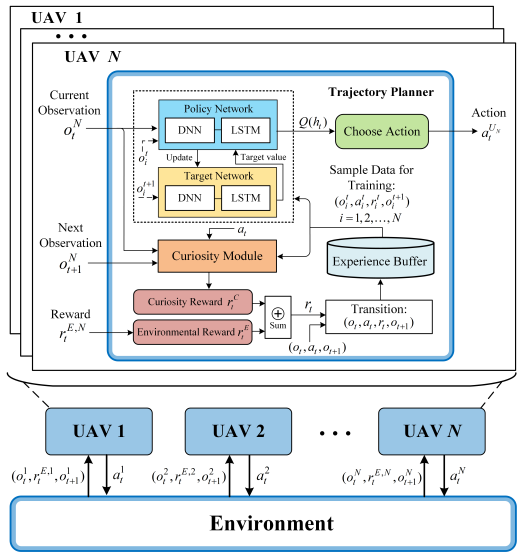

Fig. 1. The architecture of proposed CCTP algorithm.

Hence, the observation information that each UAV agent obtained from environment is denoted as:

$$o = (o_{tar}, o_B, o_I, o_U). \tag{7}$$

(b) Actions: The agents execute their selected actions simultaneously. The set of actions is denoted as $\mathcal{A} = \{a_1, a_2, \cdots, a_N\}$. As the variable of flight control in this system model is UAV steering angle, the action a is defined as the value corresponding to the steering angle. For the considerations of both computing efficiency and trajectory planning precision, a discrete action space is proposed. The action space has 11 values corresponding to the steering angles sampled form $-45°$ to $45°$.

B. DRQN-based reinforcement learning for data collection

In this work, a curiosity-enhanced cooperative trajectory planning (CCTP) algorithm for multi-UAV data collection is developed. In it, a deep recurrent Q-network (DRQN) that aggregates historical observations is utilized for action policy. Then, a dual-incentive reward mechanism that integrates extrinsic environmental rewards with intrinsic curiosity reward is used for policy training, thereby enabling robust trajectory

planning in partially observable and dynamically complex environments.

As shown in Fig. 1, each agent holds a policy network that integrates a deep neural network (DNN) with a long short term memory (LSTM). The DNN extracts the features of environment from the observation input. The LSTM stores the features of previous observations and generates a feature of inferred state based on these historical observations. Lastly, an output layer estimates values of action decisions based on the feature map from LSTM. With the help of LSTM, the trajectory planner can accurately recognize the dynamically changing environment state and make precise trajectory planning.

In the UAV agent U_i, the output of DNN is denoted as:

$$\phi\left(o_t^i\right) = DNN_{U_i}\left(o_t^i\right). \tag{8}$$

The LSTM employs gating mechanisms to selectively store the historical inputs and outputs the feature map of inferred state based on the current and historical inputs, i.e.

$$h_t^i = LSTM_{U_i}\left[\phi\left(o_t^i\right), h_{t-1}^i\right]. \tag{9}$$

Finally, the policy network outputs the state-action values based on the inferred state, which is denoted as $Q\left(h_t^i\right)$. For the optimal policy, the agent choose the action that has the highest estimated state-action value, i.e.

$$a_t^i = \arg\max_a Q\left(h_t^i, a\right). \tag{10}$$

The iteration of action value network is based on temporal difference (TD) error, i.e.

$$L\left(\theta_Q^i\right) = \left[r_t^i + \gamma \max_{a'} Q\left(h_{t+1}^i, a'\right) - Q\left(h_t^i, a_t^i\right)\right]^2, \tag{11}$$

where θ_Q^i is the parameter of policy network, γ is the discount factor. As shown in Fig. 1, the DRQN has a twin neural network structure. Policy network is deployed for trajectory planning policy and target network is deployed for calculating $Q\left(h_{t+1}^i, a'\right)$ to stabilize the TD error calculation. The parameters of target network is directly copied from policy network but the updating of target network is less frequent than policy network. A experiment buffer is built to store the transition $\left(o_t^i, a_t^i, r_t^i, o_{t+1}^i\right)$ in every step. In each iteration, a batch of transitions with batch size N_B for training policy network is randomly sampled from the buffer.

C. Intrinsic curiosity reward

The complexity of trajectory planning increases due to the unknown and dynamically changing environment. The DRQN-based trajectory planner exhibits enhanced learning efficacy for optimal trajectory planning strategies when it explores the environment comprehensively in training process. Hence, an ICM is deployed in each agent to motivate the agent to explore the environment for the globally optimal solution by giving the agent an curiosity reward in the process of training the trajectory planner. The ICM estimates the curiosity reward based on the prediction error of the next observation which reflects the novelty of experiencing an environment state transition [23].

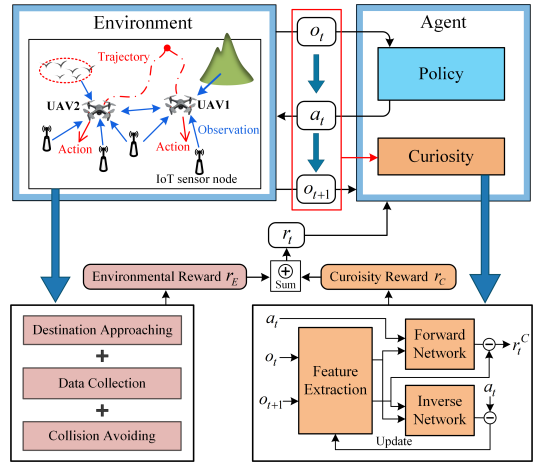

Fig. 2. The framework of intrinsic curiosity reward mechanism.

As shown in Fig. 2, within the ICM deployed in the UAV agent U_i, a twin of neural networks extract features of input observations in two steps which are denoted as $\phi_C\left(o_t^i\right)$ and $\phi_C\left(o_{t+1}^i\right)$. A forward network predicts the next observation based on the current observation and conducted action, i.e.

$$\hat{\phi_C}\left(o_{t+1}^i\right) = f_F^i\left(o_t^i, a_t^i\right). \tag{12}$$

The curiosity reward is the prediction error of the forward network, i.e.

$$r_C^i = \left[\hat{\phi_C}\left(o_{t+1}^i\right) - \phi_C\left(o_{t+1}^i\right)\right]^2. \tag{13}$$

The forward network is updated by the back propagation of state prediction error. With each iteration, the state prediction error under current input is minimized, leading to an attenuation of the curiosity toward previously experienced environment state transition.

To evaluate the qualities of feature extraction networks, an inverse network is introduced to predict the action based on the observations in adjacent two steps, i.e.

$$\hat{a}_t^i = f_I^i\left(\phi_C\left(o_t^i\right), \phi_C\left(o_{t+1}^i\right)\right). \tag{14}$$

The feature extraction networks are updated together with the inverse network by the back propagation of action prediction error which is denoted as:

$$\delta\left(a_t^i\right) = L_I\left(a_t^i, \hat{a}_t^i\right), \tag{15}$$

where the L_I is the loss function. For the discrete action space, a cross-entropy loss function is deployed. Finally, the r_C^i is propagated to the DRQN, serving as a curiosity-driven reward.

D. Environmental reward functions

After the UAV agent conducts its action, it receives a reward as the feedback from the environment. To facilitate accomplishing data collection task while jointly optimizing the collected data amount and task completion time, the reward functions are designed appropriately. For the agent U_i, the received reward are described as follows:

TABLE I
THE PARAMETER SETTINGS FOR SIMULATION.

Parameter	Value	Parameter	Value	Parameter	Value
Flying height H	50m	Noise power σ^2	-90dBm	Communication range $d^I_{\max}$	50m
Flying velocity V	10m/s	Bandwidth B	1MHz	Signal transmission power P	-20dBm
Channel parameters α, β	10, 0.6	Carrier frequency f_c	2.4GHz	Reward constants K_1, K_2, K_3	-5, -3, -0.05
Reward factors α_1, α_2	0.15, 1	Reward factors α_3, α_4	0.69, 0.115	Scaling factor of curiosity β_C	1
Learning rate of DRQN	0.0001	Batch size N_B	16	Size of experience buffer L_B	80,000
Learning rate of ICM	0.0001	Discount factor γ	0.95	Iteration period of target network N_T	100

(a) Destination approaching reward: The destination approaching reward encourages the UAV flies to the destination when no uncollected IoT node is discovered, which is denoted as:

$$r_t^{E_1,i} = \begin{cases} \alpha_1 \left(d_{t-1}^{i,tar} - d_t^{i,tar} \right), & o_I^i \neq \varnothing \\ \alpha_2 \left(d_{t-1}^{i,tar} - d_t^{i,tar} \right), & \text{otherwise} \end{cases}, \tag{16}$$

where α_1 and α_2 are scaling factors that $\alpha_1 < \alpha_2$.

(b) Data collection reward: To encourage UAV find a shortest route to collect data, when the UAV is approaching to one node, it receives the approaching reward from the node. Meanwhile, to encourage the UAV to collect as much as data amount from IoT nodes, when the UAV finished data collection from one node, it receives the collected data amount reward. The reward from the IoT sensor node I_j is denoted as:

$$r_t^{i,j} = \begin{cases} \alpha_3 \left(d_{t-1}^{i,j} - d_t^{i,j} \right), & \varphi_c^{I_j} = 0 \\ \alpha_4 D_c^{i,j}, & \varphi_c^{I_j} = 1 \end{cases}, \tag{17}$$

where $\varphi_c^{I_j}$ is the collection state of node I_j and value 1 denotes the data collection is finished, $D_{i,j}^C$ is the collected data amount from node I_j by the UAV U_i and α_3, α_4 are scaling factors. To jointly optimize the collection sequence and trajectory, the data collection reward r_{E_2} is the sum of the reward components of the observed IoT nodes, i.e.

$$r_t^{E_2,i} = \sum_{j \in \mathcal{I}_i} r_t^{i,j}, \tag{18}$$

where $\mathcal{I}_{U_i}$ is the set of IoT sensor nodes that observed by the UAV U_i.

(c) Collision reward: This reward assumes a negative scalar value to penalize suboptimal actions causing UAV's proximity to obstacles and adjacent UAVs, i.e.

$$r_t^{E_3,i} = \begin{cases} K_1, & d_t^{i,m} < d_{\min}^B \\ K_2, & d_t^{i,n} < d_{\min}^U \\ K_3, & d_t^{i,n} > d_{\max}^U \\ 0, & \text{otherwise} \end{cases}, \tag{19}$$

where K_1, K_2, K_3 are negative constants, $d_t^{i,m}$ is the distance between UAV U_i and the surface of nearest obstacle B_m, $d_{\min}^B$ is the minimum distance to the obstacle surface, $d_t^{i,n}$ is the distance between U_i and the nearest adjacent UAV U_n, $d_{\min}^U$ is the minimum safety distance between adjacent UAVs and $d_{\max}^U$ is the maximum communication maintaining distance between adjacent UAVs.

Finally, the external environment reward is the sum of these components, i.e.

$$r_t^{E,i} = r_t^{E_1,i} + r_t^{E_2,i} + r_t^{E_3,i}. \tag{20}$$

The proposed algorithm combines the extrinsic environmental reward with the intrinsic curiosity reward to formulate the agent's feedback after action execution, i.e.

$$r_t^i = r_t^{E,i} + \beta_C \cdot r_t^{C,i}, \tag{21}$$

where β_C is the scaling factor of curiosity reward.

IV. EXPERIMENT RESULTS

The multi-UAV data collection simulation is realized in Python platform and the neural networks are constructed using PyTorch. The environment region is a $1000 \times 1000\text{m}^2$ area, where 3 UAVs jointly collect the data of IoT sensor nodes. The positions of IoT sensor nodes and obstacles are stochastically assigned. The data storage capacity in IoT nodes is randomly configured with an upper bound of 6MB. The obstacles are set as cylinders with radius randomly assigned within the interval of 1m to 5m. Furthermore, the moving obstacles have stochastically generated motion trajectories. The time limitation of data collection task is 400s. The neural networks in DRQN is formed by 2 fully-connected layers and one LSTM layer. Each layer has 256 neurons. The neural networks in ICM is formed by 3 fully-connected layers. Each layer in forward network and feature extraction network has 256 neurons and each layer in inverse network has 512 neurons. The remaining parameters are specified as detailed in Table I.

For comparison, two benchmark algorithms are used: (1) One is DRQN algorithm that is same with our algorithm but without curiosity reward, which is marked as "DRQN". (2) The other is the D3QN-based algorithm proposed in [9], which is marked as "D3QN". For performance evaluation, some metrics are defined: (1) Success rate, which is the ratio of the number of successful tasks to the number of all data collection tasks. The successful task is that all the UAVs arrive at the destination. (2) Data collection rate, which is the ratio of the amount of collected data to the amount of all the stored data in successful tasks. (3) Average time, which is the average task completion time in successful tasks.

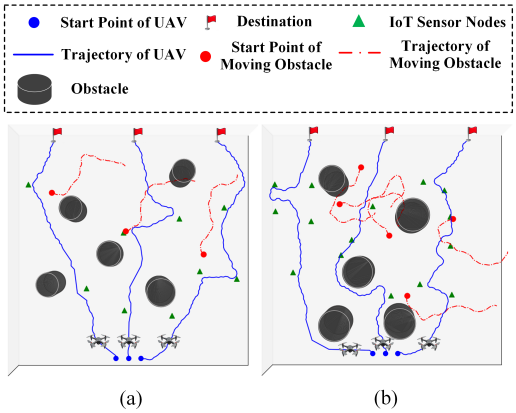

Fig. 3. The UAV trajectories in different numbers of IoT sensor nodes: (a) 9 IoT sensor nodes; (b) 15 IoT sensor nodes.

A. Data collection performance

We conduct the test in which the UAVs finish the data collection in varying numbers of IoT sensor nodes to verify the data collection performance of our algorithm. The environment in this test has 5 static obstacles, 5 moving obstacles and 6 to 24 IoT sensor nodes. As shown in Fig. 3, each UAV has a smooth trajectory, even though its work load increases.

The performance results are shown in Fig. 4. The success rate of CCTP is similar to that of DRQN, but the success rate of D3QN is about 15% lower than that of the other two algorithms when there are 15 IoT sensor nodes. The reason is that the reward function of D3QN is only related to the collected data, making D3QN can not find the best sequence and trajectory to the IoT sensor nodes and its flight time exceeds the maximum time when the number of IoT sensor nodes increases. All the algorithms have a data collection rate of over 99% and the data collection rate of CCTP is 1% higher than that of DRQN and 7% higher than that of D3QN when there are 15 IoT sensor nodes. The reason is that the CCTP has stable collection capability when the IoT node distribution is scattered. The average time of CCTP is 6.68% shorter than that of DRQN and 15.94% shorter than that of D3QN when there are 18 IoT sensor nodes, because the intrinsic curiosity reward in CCTP promotes each UAV agent to find a shorter path.

B. Obstacle avoiding performance

The obstacle avoiding performance is tested in the environment with varying numbers of moving obstacles because avoiding moving obstacles is more difficult than avoiding static obstacles. The environment in this test has 5 static obstacles, 15 IoT sensor nodes and 3 to 7 moving obstacles. The performance results are shown in Fig. 4. The success rate of CCTP is similar to that of DRQN and 12% higher than that of D3QN. As all the algorithms exhibit average time that is far from the maximum, the main contribution to the failure of task is collision. Hence, the obstacle avoiding ability of CCTP is similar to DRQN and the obstacle avoiding ability of D3QN is lower than that of the other two algorithms. Although the UAVs have to handle increasing moving obstacles, all the algorithms have the steady data collection rate compared to Fig. 4 (b) in 15 IoT sensor nodes. The average time of CCTP is 5.50% shorter than that of DRQN and 15.10% shorter than that of D3QN when there are 7 moving obstacles. The similar average time performance of obstacle avoiding test to the data collection test proves that the CCTP has the enhancement on trajectory planning ability compared to DRQN and D3QN.

V. Conclusion

This paper developes a curiosity-enhanced cooperative trajectory planning (CCTP) algorithm to address the multiple UAV data collection in complex and dynamic environment. The DRQN-based trajectory control policy enables effective decision-making for UAVs in partially observable and dynamic environment. The dual-incentive reward mechanism enhances the performance of the trained policy. Experiment results show that the proposed algorithm has better performance on planning a shorter flight trajectory to collect most of data in IoT sensor nodes. As the flight control of UAV is in horizonal plane in this paper, our future work will consider realizing the 3-D path planning.

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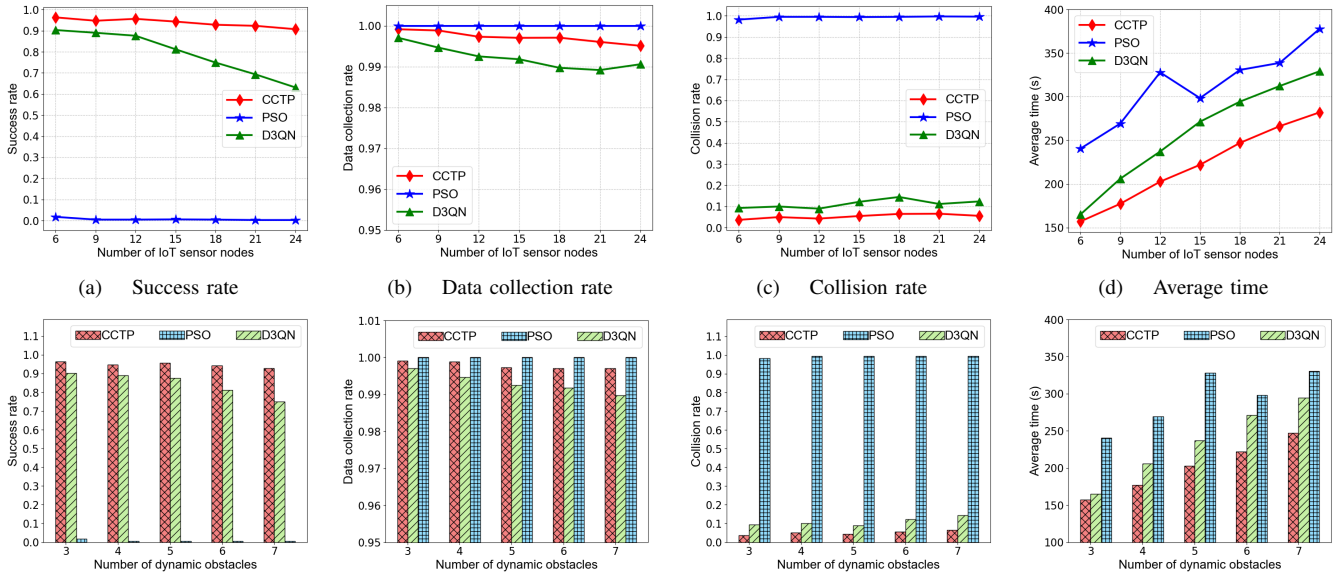

(a) Success rate (b) Data collection rate (c) Collision rate (d) Average time

(e) Success rate (f) Data collection rate (g) Collision rate (h) Average time

Fig. 4. Data collection and collision avoidance performance: (a) - (d) number of IoT sensor nodes changes; (e) - (h) number of moving obstacles varies.

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